**First-order gain model: Kalman filter, $y_2$ only, the estimate as a weighted average**

# 1 Model

**State equation**

$$\delta x_1[k+1] = \delta x_2[k]$$
$$\delta x_2[k+1] = \delta x_3[k]$$
$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k]$$
$$a_x[k+1] = a_x[k]$$

$$F_{dx}[k] = f_{dx}[k] + (1-\lambda_c)\sum_{i=1}^{d} f_{dx}[k-i] \;\approx\; \big(1 + (1-\lambda_c)d\big)\,\frac{\Delta x_d[k]}{\hat{a}_x[k]} \qquad \text{(constant velocity)}$$

**Output equation**

$$y_1[k] = \delta x_1[k] + n_x[k], \qquad R_1 = \mathrm{Var}(n_x)$$
$$y_2[k] = a_x[k] + n_a[k], \qquad R_2[k] = \mathrm{Var}(n_a[k])$$

**Estimation error and output error**

$$\mathbf{e}[k] = \begin{bmatrix} e_{\delta x_1}[k] \\ e_{\delta x_2}[k] \\ e_{\delta x_3}[k] \\ e_{a_x}[k] \end{bmatrix} = \begin{bmatrix} \delta x_1[k] - \delta\hat{x}_1[k] \\ \delta x_2[k] - \delta\hat{x}_2[k] \\ \delta x_3[k] - \delta\hat{x}_3[k] \\ a_x[k] - \hat{a}_x[k] \end{bmatrix}, \qquad \begin{aligned} e_{y_1}[k] &= y_1[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + n_x[k] \\ e_{y_2}[k] &= y_2[k] - \hat{a}_x[k] = e_{a_x}[k] + n_a[k] \end{aligned}$$

**Matrices**

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\mathbf{Q} = \mathrm{diag}(0,\, 0,\, Q_{33},\, Q_{44})$, $\mathbf{R}[k] = \mathrm{diag}(R_1,\, R_2[k])$, first order: $Q_{44} = 0$

# 2 Estimator and Kalman filter

## 2.1 Estimator (model one step ahead, plus gain times output error)

$$\delta\hat{x}_1[k+1] = \delta\hat{x}_2[k] + \ell_{21}[k]\, e_{y_1}[k] + \ell_{22}[k]\, e_{y_2}[k]$$
$$\delta\hat{x}_2[k+1] = \delta\hat{x}_3[k] + \ell_{31}[k]\, e_{y_1}[k] + \ell_{32}[k]\, e_{y_2}[k]$$
$$\delta\hat{x}_3[k+1] = \lambda_c\, \delta\hat{x}_3[k] + \big(\lambda_c\, \ell_{31}[k] - F_{dx}[k]\, \ell_{41}[k]\big)\, e_{y_1}[k] + \big(\lambda_c\, \ell_{32}[k] - F_{dx}[k]\, \ell_{42}[k]\big)\, e_{y_2}[k]$$
$$\hat{a}_x[k+1] = \hat{a}_x[k] + \ell_{41}[k]\, e_{y_1}[k] + \ell_{42}[k]\, e_{y_2}[k]$$

$$\mathbf{L}[k] = \begin{bmatrix} \ell_{11}[k] & \ell_{12}[k] \\ \ell_{21}[k] & \ell_{22}[k] \\ \ell_{31}[k] & \ell_{32}[k] \\ \ell_{41}[k] & \ell_{42}[k] \end{bmatrix} \qquad (\text{row } i = \text{state } i, \quad \text{column } 1 = y_1, \quad \text{column } 2 = y_2)$$

## 2.2 Error dynamics

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\, \big(\mathbf{I} - \mathbf{L}[k]\, \mathbf{H}\big)\, \mathbf{e}[k] - \mathbf{F}_e[k]\, \mathbf{L}[k] \begin{bmatrix} n_x[k] \\ n_a[k] \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -\varepsilon[k] \\ 0 \end{bmatrix}$$

## 2.3 Kalman filter (every step, in this order)

$$\mathbf{L}[k] = \mathbf{P}^f[k]\, \mathbf{H}^T \big(\mathbf{H}\, \mathbf{P}^f[k]\, \mathbf{H}^T + \mathbf{R}[k]\big)^{-1}$$
$$\mathbf{P}[k] = \big(\mathbf{I} - \mathbf{L}[k]\, \mathbf{H}\big)\, \mathbf{P}^f[k]$$
$$\mathbf{P}^f[k+1] = \mathbf{F}_e[k]\, \mathbf{P}[k]\, \mathbf{F}_e^T[k] + \mathbf{Q}$$

$\mathbf{P}^f[k] = \mathrm{Cov}\big(\mathbf{e}[k]\big)$ before the readings of step $k$, $\quad$ $\mathbf{P}[k]$ = after them, $\quad$ $\mathbf{P}^f[0]$ = initial value

# 3 Two tracks

## 3.1 Track 1: 2.3 (confidence and gain)

$\mathbf{P}^f[0] \xrightarrow{\mathbf{F}_e,\ \mathbf{H},\ \mathbf{Q},\ \mathbf{R}} \mathbf{L}[0],\ \mathbf{P}[0],\ \mathbf{P}^f[1] \longrightarrow \mathbf{L}[1],\ \mathbf{P}[1],\ \mathbf{P}^f[2] \longrightarrow \cdots$ ($y_1,\ y_2$ do not appear)

## 3.2 Track 2: 2.1 (estimate)

$\hat{a}_x[k] \xrightarrow{y_1[k],\ y_2[k] \text{ through } e_{y_1}[k],\ e_{y_2}[k];\ \ \ell_{41}[k],\ \ell_{42}[k] \text{ from track 1}} \hat{a}_x[k+1]$

## 3.3 Where the tracks touch

$F_{dx}[k] \ni \hat{a}_x[k]$, $Q_{33} \ni \hat{a}_x$, $R_2[k] \ni \hat{a}_x[k]$ (track 1 reads track 2 only through these)

# 4 $y_2$ only, $Q_{44} = 0$: the gain row step by step

## 4.1 2.3 with $\mathbf{H} = [\,0\;\;0\;\;0\;\;1\,]$, $\mathbf{R} = R_2[k]$ (the $a_x$ element)

$$\ell_{42}[k] = \frac{P^f_{aa}[k]}{P^f_{aa}[k] + R_2[k]}$$

$$P_{aa}[k] = \big(1 - \ell_{42}[k]\big)\, P^f_{aa}[k] = \frac{R_2[k]\, P^f_{aa}[k]}{P^f_{aa}[k] + R_2[k]}$$

$$P^f_{aa}[k+1] = P_{aa}[k] + Q_{44} = P_{aa}[k]$$

## 4.2 2.1, the $a_x$ row

$$\begin{aligned}\hat{a}_x[k+1] &= \hat{a}_x[k] + \ell_{42}[k]\,\big(y_2[k] - \hat{a}_x[k]\big)\\ &= \big(1 - \ell_{42}[k]\big)\,\hat{a}_x[k] + \ell_{42}[k]\, y_2[k]\end{aligned}$$

## 4.3 Substitute back, $k = 0, 1, 2$

$$\hat{a}_x[1] = (1 - \ell_{42}[0])\,\hat{a}_x[0] + \ell_{42}[0]\, y_2[0]$$

$$\begin{aligned}\hat{a}_x[2] &= (1 - \ell_{42}[1])\,\hat{a}_x[1] + \ell_{42}[1]\, y_2[1]\\ &= (1 - \ell_{42}[1])(1 - \ell_{42}[0])\,\hat{a}_x[0]\\ &\quad + (1 - \ell_{42}[1])\,\ell_{42}[0]\, y_2[0]\\ &\quad + \ell_{42}[1]\, y_2[1]\end{aligned}$$

$$\begin{aligned}\hat{a}_x[3] &= (1 - \ell_{42}[2])\,\hat{a}_x[2] + \ell_{42}[2]\, y_2[2]\\ &= (1 - \ell_{42}[2])(1 - \ell_{42}[1])(1 - \ell_{42}[0])\,\hat{a}_x[0]\\ &\quad + (1 - \ell_{42}[2])(1 - \ell_{42}[1])\,\ell_{42}[0]\, y_2[0]\\ &\quad + (1 - \ell_{42}[2])\,\ell_{42}[1]\, y_2[1]\\ &\quad + \ell_{42}[2]\, y_2[2]\end{aligned}$$

## 4.4 General step $k$

$$\hat{a}_x[k] = \prod_{i=0}^{k-1} \big(1 - \ell_{42}[i]\big)\;\hat{a}_x[0] + \sum_{j=0}^{k-1} c_{k,j}\, y_2[j], \qquad c_{k,j} = \ell_{42}[j] \prod_{i=j+1}^{k-1} \big(1 - \ell_{42}[i]\big)$$

### 4.5 The products telescope (4.1)

$$1-\ell_{42}[i]=\frac{P_{aa}[i]}{P^f_{aa}[i]}=\frac{P^f_{aa}[i+1]}{P^f_{aa}[i]},\qquad \ell_{42}[j]=\frac{P^f_{aa}[j]}{P^f_{aa}[j]+R_2[j]}=\frac{P_{aa}[j]}{R_2[j]}$$

$$\prod_{i=j+1}^{k-1}\big(1-\ell_{42}[i]\big)=\frac{P^f_{aa}[j+2]}{P^f_{aa}[j+1]}\cdot\frac{P^f_{aa}[j+3]}{P^f_{aa}[j+2]}\cdots\frac{P^f_{aa}[k]}{P^f_{aa}[k-1]}=\frac{P^f_{aa}[k]}{P^f_{aa}[j+1]}=\frac{P^f_{aa}[k]}{P_{aa}[j]}$$

$$c_{k,j}=\frac{P_{aa}[j]}{R_2[j]}\cdot\frac{P^f_{aa}[k]}{P_{aa}[j]}=\frac{P^f_{aa}[k]}{R_2[j]}$$

$$\prod_{i=0}^{k-1}\big(1-\ell_{42}[i]\big)=\frac{P^f_{aa}[k]}{P^f_{aa}[0]}$$

$$\frac{1}{P^f_{aa}[k+1]}=\frac{P^f_{aa}[k]+R_2[k]}{R_2[k]\,P^f_{aa}[k]}=\frac{1}{P^f_{aa}[k]}+\frac{1}{R_2[k]}\;\Rightarrow\;\frac{1}{P^f_{aa}[k]}=\frac{1}{P^f_{aa}[0]}+\sum_{j=0}^{k-1}\frac{1}{R_2[j]}$$

### 4.6 The estimate is a weighted average of the readings

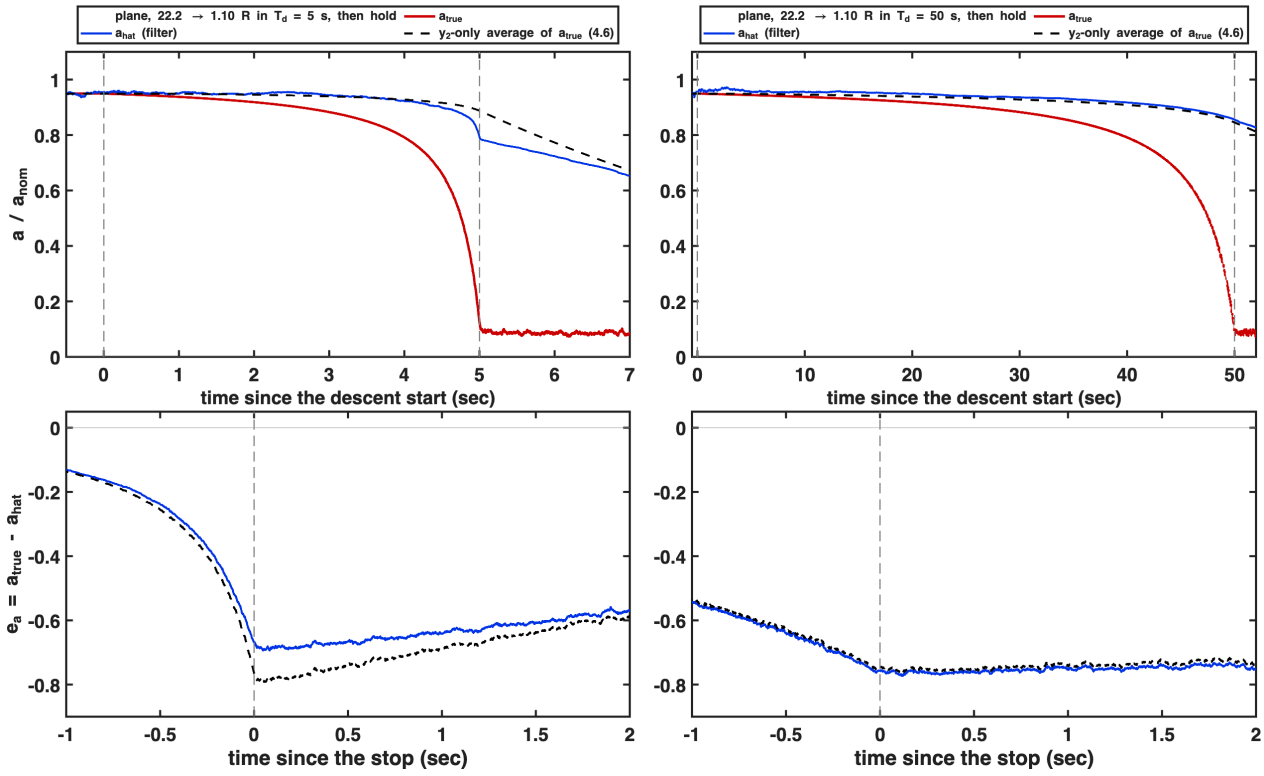
$$\boxed{\hat a_x[k]=\frac{\dfrac{1}{P^f_{aa}[0]}\,\hat a_x[0]+\displaystyle\sum_{j=0}^{k-1}\frac{1}{R_2[j]}\,y_2[j]}{\dfrac{1}{P^f_{aa}[0]}+\displaystyle\sum_{j=0}^{k-1}\frac{1}{R_2[j]}}}$$

$R_2$ constant: $\hat a_x[k]=\dfrac{m\,\hat a_x[0]+\sum_{j=0}^{k-1}y_2[j]}{m+k},\quad m=\dfrac{R_2}{P^f_{aa}[0]}$

### 4.7 Readings are true gain plus noise

$$y_2[j]=a_x[j]+n_a[j]\;\Rightarrow\;\hat a_x[k]=\frac{\dfrac{1}{P^f_{aa}[0]}\,\hat a_x[0]+\displaystyle\sum_{j=0}^{k-1}\frac{1}{R_2[j]}\,a_x[j]}{\dfrac{1}{P^f_{aa}[k]}}+\frac{\displaystyle\sum_{j=0}^{k-1}\frac{1}{R_2[j]}\,n_a[j]}{\dfrac{1}{P^f_{aa}[k]}}$$

**4.7: plane, 22.2 → 1.10 R in $T_d = 5$ s vs $50$ s, then hold; dashed = first term of 4.7 (initial estimate and true gains, weights $1/R_2[j]$)**



# 5 $y_1$ added

## 5.1 2.3, third line, the $a_x$ column of $\mathbf{P}^f$

$$
\begin{aligned}
P^f_{1a}[k+1] &= P_{2a}[k] \\
P^f_{2a}[k+1] &= P_{3a}[k] \qquad\qquad P^f_{1a} \;\to\; -\frac{F_{dx}}{1-\lambda_c}\,P_{aa} \quad (F_{dx} \text{ constant}) \\
P^f_{3a}[k+1] &= \lambda_c\,P_{3a}[k] - F_{dx}[k]\,P_{aa}[k]
\end{aligned}
$$

## 5.2 2.3, first line, both outputs

$$
\begin{aligned}
D[k] &= \big(P^f_{11}[k]+R_1\big)\big(P^f_{aa}[k]+R_2[k]\big) - \big(P^f_{1a}[k]\big)^2 \\
\ell_{41}[k] &= \frac{P^f_{a1}[k]\,R_2[k]}{D[k]}, \qquad \ell_{42}[k] = \frac{P^f_{aa}[k]\big(P^f_{11}[k]+R_1\big) - \big(P^f_{a1}[k]\big)^2}{D[k]}
\end{aligned}
$$

## 5.3 2.3, second line: the weight of one step

$$
\begin{aligned}
\frac{1}{P_{aa}[k]} &= \frac{1}{P^f_{aa}[k]} + \frac{1}{R_2[k]} + \frac{\big(P^f_{1a}[k]/P^f_{aa}[k]\big)^2}{P^f_{11}[k] - \big(P^f_{1a}[k]\big)^2/P^f_{aa}[k] + R_1} \\
&\approx \frac{1}{P^f_{aa}[k]} + \frac{1}{R_2[k]} + \frac{1}{P^f_{11}[k]+R_1}\left(\frac{F_{dx}[k]}{1-\lambda_c}\right)^2, \qquad \frac{F_{dx}}{1-\lambda_c} \propto \frac{\Delta x_d}{\hat{a}_x} \propto \frac{1}{T_d\,\hat{a}_x}
\end{aligned}
$$